Ahmed Asim Abdelhamid

Lead Site Reliability Engineer

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Profile

Experienced SRE/DevOps Engineer with a strong focus on improving processes and boosting system efficiency. Skilled in Linux and Unix operating systems, with a proven track record in implementing automation solutions. Passionate about innovating in infrastructure management. Seeking a challenging position in a forward-thinking environment to apply my technical expertise and contribute to the success of the organization.

Education

Bachelor of Engineering in Communications and Electronics Jan 2015

New Damietta Institute New Damietta, Egypt

Grade: Very Good

Professional Experience

Lead Site Reliability Engineer Jun 2023 – Present

Hala Riyadh, KSA

- Led a team of 8 SRE engineers, overseeing day-to-day operations, mentoring, and driving infrastructure reliability initiatives across the organization.
- Deployed HashiCorp Vault in HA mode and migrated 130+ microservices from plaintext secrets to dynamic secret management using External Secrets Operator and Vault Reloader, significantly improving the organization's security posture.
- Adopted GitOps methodology by deploying Argo CD and migrating 130+ microservices from a Helmfile-based deployment approach to a fully declarative Argo CD workflow, improving deployment consistency and auditability.
- Designed and implemented a hub-and-spoke network architecture on Oracle Cloud Infrastructure, deploying dedicated firewalls for east-west and north-south traffic inspection and enforcement.
- Built CI/CD pipelines for Node.js and .NET microservices, standardizing the build, test, and deployment lifecycle across the engineering organization.
- Implemented Kong API Gateway as the centralized ingress layer, enabling unified rate limiting, authentication, and traffic management across services.

- Provisioned a full Disaster Recovery site on the OCI Riyadh region using Terraform IaC, ensuring infrastructure reproducibility and rapid failover capability.
- Implemented Disaster Recovery for MongoDB Ops Manager, enabling cross-region backup orchestration and recovery readiness.
- Managed and maintained Patroni-based PostgreSQL clusters, ensuring high availability, automated failover, and reliable database operations for critical workloads.
- Implemented and validated a comprehensive Disaster Recovery (DR) solution on Alibaba Cloud, successfully executing a 7-day (5 business days) full production workload transition from OCI. Demonstrated system resilience through seamless operations during the DR period and achieved smooth failback to the primary site, with minimal downtime during both failover and failback processes.
- Deployed Cross-Site Replication: Managed the deployment of Cross-Site Replication for Percona MongoDB Clusters, handling over 20 TB of critical data on Alibaba Cloud for DR purposes.
- Introduced Robust VPN Solution: Led the implementation of Tailscale as a secure VPN solution, utilizing Kubernetes deployments across multiple subnet routers.
- Streamlined Access Control: Improved access control management using Teleport, ensuring secure and efficient access protocols.
- Utilized Azure Active Directory: played a key role in utilizing Azure Active Directory for effective identity and access management.
- Enhanced Security Measures: Strengthened infrastructure security by collaborating with Cloudflare WAF to protect against potential threats.
- Managed Oracle Cloud Infrastructure: Oversaw HALA Infrastructure on Oracle Cloud using Terraform, ensuring scalability, reliability, and cost-effectiveness.
- Developed Incident Response Service: Created a Golang service to trigger PagerDuty Critical Alerts directly from Slack via a custom Slack command, improving incident response times.
- Optimized Service Mesh: Configured and optimized Istio Service Mesh, enhancing service discovery, security, and observability within the infrastructure.
- MongoDB Enterprise: Migration of 8 high-volume clusters, encompassing 20TB of critical production data, from Percona MongoDB open-source to MongoDB Enterprise with Ops Manager. Successfully implemented replication for the Disaster Recovery (DR) site, enhancing system resilience and business continuity.

Sr. DevOps Engineer Jul 2021 – May 2023
Takamol Holding Riyadh, Saudi Arabia

- Infrastructure Management: Oversee infrastructure on STC Cloud and Oracle Cloud Infrastructure, ensuring optimal performance and reliability.

- Kubernetes Clusters: Direct the management and enhancement of Kubernetes Clusters, particularly within the Oracle Kubernetes Engine environment.
- Application Management with Helm and ArgoCD: Utilize Helm charts and ArgoCD to manage Kubernetes applications efficiently, following GitOps methodology for streamlined deployment and version control.
- Auto-Scaling: Implement Kubernetes cluster auto-scaling in production environments to dynamically adjust resource allocation based on workload demands.
- Horizontal Pod Autoscaling: Enable Horizontal Pod Autoscaling for Kubernetes applications across all environments, enhancing scalability and resource utilization.
- Secure Access: Implement Pritunl VPN for secure access to pre-production and internal applications, prioritizing data security and access control.
- Web Application Security: Deploy Nginx App Protect WAF and DOS solutions to protect applications against web attacks and DDoS threats, ensuring robust security measures.
- Elasticsearch Management: Manage Elasticsearch clusters to efficiently cache critical governmental data in production environments, optimizing data retrieval and accessibility.
- Seamless Upgrades: Conduct rolling upgrades for Elasticsearch clusters with seamless transitions and zero downtime, ensuring uninterrupted service availability.
- CI/CD Pipelines: Facilitate the deployment of microservices through GitLab CI/CD pipelines, streamlining the software delivery process.
- Access Control and Secret Management: Collaborate with Teleport for access control management and Vault for secure secret management, bolstering access control and data security protocols.
- Containerization: Utilize Docker and Docker Compose for the containerization and orchestration of applications, ensuring consistency and portability across environments.
- Production Issue Resolution: Resolve production issues at the infrastructure level for Linux-based systems, ensuring system stability and reliability.
- Network Security: Leverage F5 Firewall for VPN access, implementing robust network security measures.
- Development Support: Provide critical support to the development team for the timely resolution of high-priority issues, fostering collaborative problem-solving and incident response.
- DNS Management: Manage external DNS services with Cloudflare, optimizing domain resolution and network performance.
- Database Optimization: Engage with Postgres Database and Heimdall Data Proxy to optimize database performance and streamline data access.

- On-Call Support: Participate in on-call shifts and promptly respond to high-critical alerts 24/7, ensuring continuous system monitoring and availability.

Sr. Systems Administrator Jun 2020 – Jul 2021
National Financing Services Company Riyadh, Saudi Arabia

- Oversee the management and maintenance of servers, storage, and virtual infrastructure, encompassing VMware (vCenter, vSphere) and DELL hardware (PowerEdge R7525, PowerEdge R640, Storage PowerVault ME4024).
- Monitor and manage diverse network infrastructures, including PaloAlto and Pfsense firewalls, as well as Cisco switches.
- Maintain and administer Active Directories, file sharing, IIS, LDAP, and DNS infrastructure for both Datacenter and Cloud environments.
- Manage Linux Ubuntu Servers across Datacenter and Cloud environments, ensuring optimal performance and security.
- Implement stringent access controls and firewall configurations to enhance security across both Datacenter and Cloud environments.
- Administer and monitor VPN connections using OpenVPN and Global Protect for seamless connectivity across sites.
- Administer Cloud Environment (STC cloud), managing OpenStack backend, virtual machines, firewalls, networking, Windows & Linux Servers, and volumes.
- Perform operational tasks including monitoring, maintenance, and problem resolution to maintain a stable network and systems environment.
- Install, configure, and maintain Windows and Linux systems with necessary application software.
- Manage integration between LDMS, Brightware, and SADAD systems, ensuring seamless data flow and functionality.
- Administer domain and subdomains, optimizing organization and accessibility.
- Manage Avaya Systems (IPOFFICE, EMS, SBCE, ACCS, AWFOS), configuring Avaya Agents' hard and softphones.
- Administer Microsoft SQL Databases, ensuring data integrity and availability.
- Install, configure, manage, and secure SFTP server used by local Saudi banks to share sensitive documents with NFSC.
- Provide technical support for LDMS vendor engineers, addressing application, database, and server-related issues.
- Troubleshoot and resolve technical issues on SRC laptops, including FortiGate VPN, email, network share, and connectivity issues.
- Provide technical support for NFSC employees, addressing various technical requirements and issues.
- Implement and configure scheduled backups using Restic to backup NFSC Oracle database, WebLogic app, and other critical documents.
- Install and configure monitoring tools such as ManageEngine, Prometheus, and monit for comprehensive system monitoring.

- Collaborate with vendors to provide backup solutions and SOC for NFSC in alignment with organizational requirements, coordinating with the procurement department as needed

DevOps Engineer Dec 2019 – Jun 2020
SlashTech Dubai, United Arab Emirates

- Develop infrastructure as code using Terraform and CloudFormation, ensuring efficient and scalable deployment of resources.
- Craft Ansible playbooks to remotely configure environments on AWS cloud, streamlining configuration management processes.
- Create Docker images from scratch and containerize existing applications, optimizing deployment and resource utilization.
- Establish logging infrastructure by deploying and configuring ELK stack based on Docker, and configuring Filebeat to transmit Nginx logs to Elasticsearch for centralized monitoring and analysis.
- Implement monitoring solutions by deploying and configuring Prometheus as a data source and Grafana for visualizing system metrics, enhancing observability across the infrastructure.
- Configure alerting mechanisms using Alert Manager to send notifications to Slack, ensuring timely response to critical events.
- Develop Jenkins pipelines for projects, facilitating Continuous Integration/Continuous Deployment (CI/CD) processes and accelerating software delivery cycles.
- Collaborate with Code Deploy and Code Pipeline services to implement and streamline CI/CD workflows, optimizing development and deployment processes.
- Manage MongoDB instances, including the creation of MongoDB replication for three nodes on AWS, ensuring data availability and reliability.
- Utilize AWS Trusted Advisor service to optimize performance, reduce costs, enhance security, and automate processes within AWS environments.
- Oversee the management and maintenance of environments on AWS, ensuring optimal performance, reliability, and security.
- Provide technical support to SlashTech clients, addressing and resolving technical issues in a timely and efficient manner.

Product Technology Engineer (NA RSC – BSS – CBS Domain) Oct 2018 – Nov 2019
Huawei Technologies Cairo, Egypt

- MTN South Sudan Project CBS 3.3:
  - Oversaw the cloning of applications (BMPAPP, CBPAPP, GFEP, etc.) from Primary (PR) to Disaster Recovery (DR) environments.
  - Led the commissioning and integration between CBS System's internal Network Elements (NEs) and External Modules.
  - Provided ongoing support post-launch, including monitoring system performance, troubleshooting issues, and delivering customer support.
- Airtel Congo B CBS 5.6 Project:

- Conducted thorough inspections and rectifications of all CBS Modules (OS, Service, DB) to ensure optimal functionality.
- Managed the inspection and rectification processes for E9000, Eudemon Huawei Firewall, Network Switches, and Huawei Storage v5500.

- Tigo Senegal CBS 5.6 Project:
  - Oversaw inspection and rectification activities for all CBS Modules (OS, Service, DB) to maintain system integrity and performance.
  - Managed the inspection and rectification of E9000, Eudemon Huawei Firewall, Network Switches, and Huawei Storage v5500 components.

Linux and Unix Systems Engineer Aug 2016 – Oct 2018

Code95 Ismailia, Egypt

- Provided 24/7 technical support to Code95 clients, ensuring adherence to Service Level Agreements (SLA) and resolving issues promptly.
- Conducted troubleshooting of Linux Web server issues, including Apache, Nginx, MySQL, PHP, and Varnish, to maintain server performance and reliability.
- Addressed Linux Mail Server issues on platforms such as cPanel, Postfix, and Zimbra, ensuring seamless email communication for clients.
- Utilized Ansible Configuration Management Tool to deploy and automate Code95 Servers, streamlining server provisioning and maintenance processes.
- Collaborated in the implementation of Continuous Integration/ Continuous Deployment (CI/CD) pipelines using DevOps tools such as GIT for version control, Jenkins for continuous integration, and Docker/Kubernetes for containerization and orchestration, fostering efficient software delivery practices.

Certificates

Certified Kubernetes Administrator (CKA) | Certified Kubernetes Application Developer (CKAD) | Udemy DevOps Nanodegree | Redhat Certified System Administrator (RHCSA)

Skills

Kubernetes, Oracle Cloud Infrastructure, Alibaba Cloud Infrastructure, ArgoCD, Helm Charts, Terraform, Nginx WAF & Nginx DOS, Gitlab, Ansible, Elasticsearch, Prometheus & Grafana & Alert manager, Amazon Web Services, MySQL, MariaDB, MongoDB, CDNs (Cloudflare – stack path), Linux Systems, MongoDB Enterprise, Percona MongoDB